

EXPERIENCE

Software Developer Engineer II, Project 990 – Bloomington, IN Jul 2025 – Present

- Achieved 8x speedup in query response time over a 5M node & 50M relationship graph database by optimizing data structures and implementing concurrent traversal techniques, which aligns with the performance goals in GPGPU operations.
- Developed and optimized microservices using .NET 5 and implemented CI/CD pipelines with GitHub Actions, reducing deployment time by 80%, demonstrating strong software engineering practices.
- Collaborated with cross-functional teams to document feature requirements and design specifications, ensuring that development aligns with hardware/software interface needs.
- Tech: C++, Python, .NET 5 (ASP.NET Core), AWS, Gremlin, Docker, Kubernetes, GitHub Actions, REST, JSON

Software Engineer, App Orchid – Hyderabad, India Apr 2022 – Aug 2023

- Led a team of 3 engineers to re-architect a scalable Machine Learning data ingestion engine, generating \$1M ARR, and acted as the primary owner for the project.
- Developed and optimized APIs for high-performance GPGPU operations using Java and integrated various machine learning models, enhancing inference speed and accuracy.
- Collaborated with cross-functional teams to implement microservices architecture with PostgreSQL and DynamoDB on AWS, improving system scalability and reliability.
- Automated build processes utilizing Azure, Docker, and Kubernetes, establishing a CI/CD pipeline to enhance software delivery quality and efficiency.
- Tech: Java, Spring Boot, PostgreSQL, DynamoDB, AWS, Docker, Kubernetes, GPGPU operations, Machine Learning, RESTful API.

Software Engineer, HDFC Bank – Bengaluru, India Jul 2020 – Apr 2022

- Developed and optimized machine learning pipelines for high-performance GPGPU operations, enhancing inference accuracy for over 50K audio calls deployed as cloud-hosted serverless services.
- Engineered batch ETL workflows in Apache Airflow, implementing fan-out/fan-in parallel processing with robust error handling to improve processing efficiency.
- Constructed efficient database schemas and transitioned from LINQ to T-SQL stored procedures, achieving a 25% performance improvement in data retrieval operations.
- Tech: C++, Python, Apache Airflow, Docker, Kubernetes, PostgreSQL, Redis, Apache Kafka, Elasticsearch, BERT. Familiar with GPGPU APIs and Machine Learning frameworks.

GPU Software Kernel Development Engineer, Qualcomm GPU Software Team – Buffalo, NY May 2024 – Aug 2024

- Developed and optimized Adreno GPU drivers for standard APIs such as OpenCL and Vulkan, enhancing performance and efficiency for GPGPU operations.
- Collaborated closely with hardware designers to improve and document hardware/software interfaces, ensuring seamless integration between software and Snapdragon chipsets.
- Accelerated Machine Learning operations for inference and training on the GPU, contributing to performance improvements and best-in-class features.
- Wrote detailed design documentation and feature descriptions, supporting external developers and OEMs in implementation and customization.
- Utilized strong C/C++ programming skills to write code for well-defined features, drivers, and APIs, enhancing overall software functionality.

Graduate Research Assistant, Prof. Prateek Sharma – Indiana University Aug 2024 – May 2025

- Architected and implemented high-performance distributed inference services in C++ and CUDA, optimizing memory layout and GPU utilization, achieving a latency reduction of 20x for Machine Learning workloads.
- Developed low-latency communication pipelines utilizing gRPC, enhancing data transfer efficiency for GPGPU applications.
- Optimized system performance through multi-threading, stream concurrency, and advanced memory management techniques to meet rigorous performance standards.
- **Publication:** MQFQ-Sticky — Fair Queueing for Serverless GPU Functions, focusing on optimizing GPU resource allocation.

PROJECTS

- GPU Software Kernel Development** | Qualcomm
- Engineering Group
- Developed and optimized high-performance GPU drivers for Adreno GPUs, enhancing GPGPU operations for Machine Learning and Image Processing.
 - Collaborated with hardware designers to improve and document hardware/software interfaces, ensuring seamless integration with the graphics software stack on Android, Linux, and Windows.
 - Implemented and maintained APIs utilizing OpenCL and Vulkan to accelerate Machine Learning operations for inference and training on the GPU.
 - Wrote comprehensive documentation and unit tests for GPU drivers and frameworks, facilitating collaboration with external developers and OEM partners.

- GPU Driver Development and Optimization** | Qualcomm GPU Software Team
- Software Engineering I
- Collaborated with cross-functional teams to develop and optimize Adreno GPU drivers for standards such as OpenCL and Vulkan, enhancing performance for Machine Learning and Image Processing tasks.
 - Contributed to the acceleration of Machine Learning operations on GPGPU, improving inference times by up to 30% through efficient kernel development and optimization.
 - Wrote detailed design documentation and feature descriptions for driver enhancements, ensuring clarity in hardware/software interface specifications.
 - Utilized strong C/C++ programming skills to implement well-defined features and APIs, driving successful project outcomes in a fast-paced environment.

SKILLS

Languages	C/C++, Java, Python, JavaScript, TypeScript, SQL
Technology	.NET Core, React, Node, Spring Boot, REST API, Flask, FastAPI, Django, Git, Jira
Data & Tools	AWS, EC2, DynamoDB, S3, SQS, Lambda, Elasticsearch, Spark, Docker, Splunk, Kafka, RabbitMQ

EDUCATION

M.S. in Computer Science, Indiana University , Bloomington, IN GPA: 3.8/4.0	Aug 2023 – May 2025
B.Tech in Aerospace, Minor in Computer Science, Indian Institute of Technology, Bombay	Jul 2016 – Jul 2020